

Answer **all** questions in the spaces provided.

0 1 . 1

The nucleus and a chloroplast of a plant cell both contain DNA.

Give **three** ways in which the DNA in a chloroplast is different from DNA in the nucleus.

[3 marks]

1 _____

2 _____

3 _____

0 1 . 2

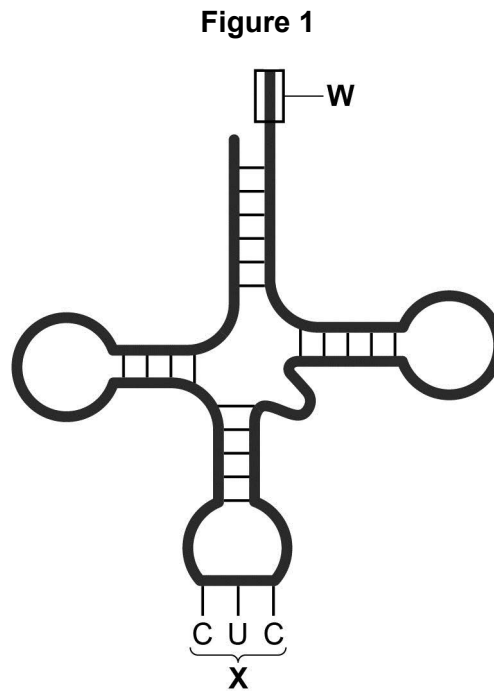
Some DNA nucleotides have the organic base thymine, but RNA nucleotides do not have thymine. RNA nucleotides have uracil instead of thymine.

Give **one** other difference between the structure of a DNA nucleotide and the structure of an RNA nucleotide.

[1 mark]



Figure 1 shows a tRNA molecule.



0 1 . 3 Name the structures labelled **W** and **X** in **Figure 1**.

[1 mark]

W _____

X _____

0 1 . 4 Not all mutations in the nucleotide sequence of a gene cause a change in the structure of a polypeptide.

Give **two** reasons why.

[2 marks]

1 _____

2 _____

Turn over ►

